

Viet K. Nguyen

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Research Interests

Security of AI agents and of autonomous cyber-physical systems. I study how attacks get into these systems, how far they travel, and how to detect and stop them, and I test this on real models and real robots.

Education

California State Polytechnic University, Pomona

M.S. in Computer Science, GPA 4.00/4.00

Pomona, CA

Aug 2026

Project: A Unified Architectural Framework and Empirical Fingerprinting of Cyber Attacks on Autonomous Cyber-Physical AI Systems. *Advisor:* Mohammad Husain.

California State Polytechnic University, Pomona

B.S. in Computer Science, GPA 3.80/4.00, summa cum laude, President's and Dean's Lists

Pomona, CA

May 2024

Publications

Peer-reviewed

- [1] **Viet K. Nguyen**, Mohammad I. Husain. "Penetration Testing of Agentic AI: A Longitudinal Security Analysis Across Models and Frameworks." *IEEE Conference on Communications and Network Security (CNS)*, 2026. Earlier version: [arXiv:2512.14860](https://arxiv.org/abs/2512.14860). Code and traces: github.com/VietNguyen705/agentic-ai-pentest.
- [2] **Viet K. Nguyen**, Kai C. Xue, Nathan Lee, Mohammad I. Husain. "SoK: A Unified Architectural Framework of Cyber Attacks on Autonomous Cyber-Physical AI Systems." *11th IEEE International Workshop on Cyber-Physical Systems Security (CPS-Sec)*, at IEEE CNS 2026.
- [3] Abdelfattah Amamra, **Viet K. Nguyen**, Adam Cheung, Sarah Acosta, Thuy Linh Pham. "Wavelet-Based IoT Device Fingerprinting." *Electronics*, 15(4):786, 2026. [doi:10.3390/electronics15040786](https://doi.org/10.3390/electronics15040786).

Preprints and under review

- [4] **Viet K. Nguyen**, Nathan Lee, Mohammad Husain. "Experimental Evaluation of Security Attacks on Self-Driving Car Platforms." [arXiv:2603.14124](https://arxiv.org/abs/2603.14124), 2026. Accepted as a short paper at SECURE 2026; withdrawn because I could not travel to present.
- [5] **Viet K. Nguyen**, Mohammad I. Husain. "An Experimental Evaluation of Multimodal Prompt Injection Attacks on Agentic AI Frameworks." [arXiv:2609.09404](https://arxiv.org/abs/2609.09404). Under review at the ACSAC Workshop on AI-Driven Cybersecurity (AIDC), 2026.

In preparation

- [6] **Viet K. Nguyen**, Mohammad I. Husain. "Agent Trust Infrastructure (ATI): A Federated Identity, Trust, Attestation, and Accountability Architecture for Cross-Vendor Autonomous AI Agents." arXiv submission in progress.
- [7] **Viet K. Nguyen**. "Security Assessment of a VR-Teleoperated Humanoid Robot: Attack Surface of the Unitree G1 Teleoperation Pipeline." Technical report, PolySec Cyber Lab; findings reported to the vendor.
- [8] Abdelfattah Amamra, **Viet K. Nguyen**, Adam Cheung, Sarah Acosta, Thuy Linh Pham. "Fourier Neural Operator-Based IoT Device Fingerprinting."

Research Experience

Security of AI Agents

Advisor: Mohammad Husain

PolySec Cyber Lab, Cal Poly Pomona

Jun 2025 – Aug 2026

- Red-teamed a seven-agent system on AutoGen and CrewAI with 13 attacks, across 2025 models and their 2026 successors (650 traces, LLM judge checked against a human audit at 94% agreement). The framework mattered as much as the model: AutoGen contained 60.9% of attacks, CrewAI 48.3%. Found *hallucinated compliance*, where an agent fakes a successful result instead of refusing [1].
- Built MMPIBench, which hides attacks in images (OCR text, overlays, EXIF, QR codes, fake interfaces) and follows each one from perception to tool call. Over 720 runs on 5 models and 6 frameworks, attacks were attempted in 12.8% of runs and completed in about 1%. The model mattered far more than the framework [5].
- Designed Agent Trust Infrastructure: a certificate authority per organization (AgentCA), runtime attestation, authorization at every tool call, and a signed hash-chained log. In simulation across 8 organizations and 6 frameworks,

it raised attack containment from 43% to 84% with no loss in benign task success, at about $174\mu s$ per request [6].

- Built an offline LLM assistant that checks each user’s role before retrieving a document and runs on one consumer GPU, for organizations that cannot send data to cloud models.
- Built a CyberGym “green agent” for the Berkeley RDI AgentX–AgentBeats competition. It scores how well AI agents find the root cause of real CVEs (ARVO, OSS-Fuzz) and write working proof-of-concept inputs.

Security of Autonomous Cyber-Physical Systems (AFRL-sponsored)

PolySec Cyber Lab, Cal Poly Pomona

Advisor: Mohammad Husain

Jan 2025 – Aug 2026

- Built a 12-layer reference architecture and attack taxonomy shared by cars, drones, humanoids, quadrupeds, ground robots and robot vacuums, so attack studies on different platforms can be compared directly [2]. Grounded it in our own on-hardware study: five attack classes (FGSM, PGD, man-in-the-middle, denial of service, projected phantom objects) on JetRacer and Yahboom self-driving cars over 150+ trials. Each attack leaves a distinct fingerprint in steering, compute cost and frame rate; PGD alone raised processing time by 3,360% and cut frame rate by 68.5% [4]. Combined both in my M.S. project.
- As the sole researcher, assessed the Unitree G1 humanoid’s VR teleoperation setup (Meta Quest 3). Showed that both the control link and the robot’s internal DDS network accept unauthenticated clients, and demonstrated interception, replay, command tampering and arm control with no operator present [7]. Also collected teleoperation demonstrations for training vision-language-action robot models.
- Pen-tested a Boston Dynamics SPOT robot and found a denial-of-service flaw (CVSS 7.5) that made it unreachable.

Machine Learning for Network Security

Cal Poly Pomona

Advisor: Abdelfattah Amamra

May 2025 – Aug 2026

- Identified IoT devices from network traffic with wavelet features, reaching 99% accuracy across 28–105 devices on three public datasets and holding 71% accuracy during a DDoS attack [3].
- Led a federated version with Fourier Neural Operators, working with two student researchers. Under extreme resolution skew between clients it keeps 0.940 accuracy where a CNN falls to 0.673.

Talks and Posters

- “Penetration Testing of Agentic AI.” Talk, IEEE CNS 2026, Newark, DE, Sep 14, 2026.
- “SoK: A Unified Architectural Framework of Cyber Attacks on Autonomous Cyber-Physical AI Systems.” Talk, CPS-Sec 2026 at IEEE CNS, Newark, DE, Sep 2026.
- “Experimental Evaluation of Security Attacks on Self-Driving Cars.” Poster, Southern California AI and Robotics Symposium (SCAIR), Oct 2025.
- “Wavelet-Based IoT Device Fingerprinting.” Poster, Cal Poly Pomona Cybersecurity Awareness Fair, Oct 2025 (2nd place of 30).

Mentoring

- Designed nine guided research projects (Jupyter notebooks with solutions) for high school students in a summer research program at Cal Poly Pomona, 2026, and held weekly two-hour office hours to help them with their projects.

Skills

PyTorch, adversarial ML, LLM agent frameworks (AutoGen, CrewAI, LangGraph, OpenAI Agents SDK, Semantic Kernel), RAG and local LLM serving, network pen testing; robot hardware: Unitree G1, Boston Dynamics SPOT, NVIDIA Jetson, Meta Quest 3.